

MANAGEMENT & PROJECT EXAM STUDY GUIDE

Covers: Management Theory • Project Management • Fayol • Mintzberg • CPM • PERT • Cost • PP •
ARR • NPV • IRR

PART 1: STUDY NOTES | **PART 2: EXAM PRACTICE**

PART 1: STUDY NOTES

1. Definition of Management

Management is the process of planning, organising, leading, and controlling an organisation's resources (people, money, materials, information) to achieve stated goals efficiently and effectively.

Key Definitions by Scholars

Scholar	Definition
Mary Parker Follett	Management is the art of getting things done through people.
Peter Drucker	Management is about doing things right; leadership is about doing the right things.
Henri Fayol	To manage is to forecast and plan, to organise, to command, to co-ordinate, and to control.
Harold Koontz	Management is the art of getting things done through and with people in formally organised groups.

Why Management Matters

- Achieves organisational goals effectively
- Optimises the use of scarce resources
- Provides direction and leadership
- Facilitates adaptation to change
- Creates a productive work environment

2. What is a Project?

A project is a temporary endeavour undertaken to create a unique product, service, or result. It has a defined beginning and end, and is carried out to meet unique goals and objectives.

Characteristics of a Project

Characteristic	Description
Temporary	Has a definite start and end date
Unique	The output is different from routine operations
Progressive	Developed in steps and continues by increments
Goal-oriented	Aimed at achieving specific objectives
Resource-constrained	Uses limited budget, people, and time
Interdependencies	Tasks are interrelated and sequenced

Project vs. Operations

Project	Operations
Temporary	Ongoing / permanent
Unique output	Repetitive output
Closes when goal met	Continues indefinitely
Example: Building a bridge	Example: Collecting tolls on bridge

The Project Management Triangle (Triple Constraint)

Every project is constrained by three factors:

- Scope (what must be done)
- Time (when it must be done)
- Cost (how much it can cost)

Changing one constraint affects the others. Quality sits at the centre.

3. Henri Fayol – Father of Modern Management

Henri Fayol (1841–1925) was a French mining engineer and management theorist. He developed the Administrative Theory of Management, based on his practical experience running large organisations. He identified 5 functions and 14 principles of management.

3.1 Fayol's 5 Functions of Management

Function	Meaning
1. Planning	Setting objectives and deciding how to achieve them
2. Organising	Arranging resources and tasks to achieve goals
3. Commanding	Directing and leading employees
4. Coordinating	Ensuring all activities work in harmony
5. Controlling	Monitoring performance and taking corrective action

Memory tip: POCCC – Plan, Organise, Command, Coordinate, Control

3.2 Fayol's 14 Principles of Management

#	Principle	Meaning
1	Division of Work	Specialisation increases efficiency
2	Authority & Responsibility	Right to give orders; must match responsibility
3	Discipline	Employees must obey rules and agreements
4	Unity of Command	One employee, one boss only
5	Unity of Direction	One plan for all activities with the same objective
6	Subordination of Individual Interest	Company goals take priority over personal goals
7	Remuneration	Fair pay for employees and employers
8	Centralisation	Degree to which decision-making is concentrated at top
9	Scalar Chain	Clear line of authority from top to bottom (hierarchy)
10	Order	Right person in the right place; materials where needed

#	Principle	Meaning
11	Equity	Fair and kind treatment of employees
12	Stability of Tenure	Low staff turnover; keep good employees
13	Initiative	Encourage employees to come up with ideas
14	Esprit de Corps	Team spirit and unity in the organisation

4. Mintzberg's Managerial Roles

Henry Mintzberg (born 1939) is a Canadian management academic. He argued that management cannot be reduced to simple functions. Instead, he observed what managers actually do and grouped their activities into 10 roles across 3 categories.

4.1 The 10 Managerial Roles

Category	Role	Description
INTERPERSONAL	Figurehead	Ceremonial duties – signing documents, attending events
INTERPERSONAL	Leader	Motivating & directing employees, hiring, training
INTERPERSONAL	Liaison	Networking with people outside the work unit
INFORMATIONAL	Monitor	Gathering internal & external information
INFORMATIONAL	Disseminator	Sharing information within the organisation
INFORMATIONAL	Spokesperson	Transmitting info to outsiders on behalf of the org
DECISIONAL	Entrepreneur	Initiating change, innovation, new projects
DECISIONAL	Disturbance Handler	Dealing with crises and unexpected problems
DECISIONAL	Resource Allocator	Distributing budgets, time, equipment, staff
DECISIONAL	Negotiator	Negotiating with stakeholders inside and outside

4.2 Fayol vs. Mintzberg Comparison

Aspect	Fayol	Mintzberg
Approach	Prescriptive (what managers should do)	Descriptive (what managers actually do)
Focus	5 functions of management	10 roles across 3 categories
View	Management as systematic process	Management as fragmented, varied activity
Time Period	Early 1900s	1970s onwards
Criticism of other	N/A	Fayol's functions are too simplistic

5. Critical Path Method (CPM)

CPM is a project scheduling technique used to determine the longest sequence of dependent tasks (the critical path) and the minimum project duration. It was developed in the late 1950s by DuPont Corporation.

5.1 Key CPM Concepts

Term	Meaning
Activity	A task that takes time and resources to complete
Event/Node	A point in time (start or finish of an activity)
Critical Path	The longest path through the network – determines project duration
Float / Slack	The amount of time an activity can be delayed without delaying the project
EST	Earliest Start Time – earliest an activity can begin
EFT	Earliest Finish Time = EST + Duration
LST	Latest Start Time – latest activity can start without delaying project
LFT	Latest Finish Time = LST + Duration
Total Float	LFT – EFT OR LST – EST (zero means activity is critical)

5.2 Forward & Backward Pass

Forward Pass (Left to Right) – Find EST & EFT

$$\text{EFT} = \text{EST} + \text{Duration}$$

Start at the first node (EST = 0). At each node, the EST = the highest EFT of all incoming activities.

Backward Pass (Right to Left) – Find LST & LFT

$$\text{LST} = \text{LFT} - \text{Duration}$$

Start at the last node (LFT = Project Duration). At each node, the LFT = the lowest LST of all outgoing activities.

5.3 Identifying the Critical Path

- Activities with Total Float = 0 are CRITICAL
- The critical path is the chain of critical activities from start to finish

- Any delay on the critical path delays the entire project
- Activities not on the critical path have float and can be delayed by that amount

5.4 Worked Example

Activities and durations:

Activity	Predecessor	Duration (days)
A	None	3
B	None	4
C	A	2
D	B	5
E	C, D	3

Forward Pass:

- A: EST=0, EFT=3
- B: EST=0, EFT=4
- C: EST=3, EFT=5
- D: EST=4, EFT=9
- E: EST=max(5,9)=9, EFT=12 → Project duration = 12 days

Backward Pass:

- E: LFT=12, LST=9
- D: LFT=9, LST=4
- C: LFT=9, LST=7
- B: LFT=4, LST=0
- A: LFT=7, LST=4

Float:

- A: Float = 7-3 = 4 (not critical)
- B: Float = 4-4 = 0 (CRITICAL)
- C: Float = 9-5 = 4 (not critical)
- D: Float = 9-9 = 0 (CRITICAL)
- E: Float = 12-12 = 0 (CRITICAL)

Critical Path: B → D → E = 12 days

6. Program Evaluation and Review Technique (PERT)

PERT is a project management tool used for scheduling, organising, and coordinating tasks within a project. Unlike CPM (which uses fixed time estimates), PERT uses THREE time estimates to account for uncertainty.

6.1 PERT Time Estimates

Estimate	Symbol	Meaning
Optimistic time	a	Best case scenario – if everything goes well
Most likely time	m	Most realistic estimate under normal conditions
Pessimistic time	b	Worst case scenario – if everything goes wrong

6.2 PERT Formulas

Expected Time (te):

$$te = (a + 4m + b) \div 6$$

This is a weighted average where the most likely time has 4x the weight.

Variance (σ^2):

$$\sigma^2 = [(b - a) \div 6]^2$$

Standard Deviation (σ):

$$\sigma = (b - a) \div 6$$

6.3 CPM vs. PERT

CPM	PERT
Uses single (deterministic) time estimate	Uses 3 time estimates (probabilistic)
Best for well-known, routine projects	Best for new, unique, uncertain projects
Focuses on cost-time trade-offs	Focuses on time uncertainty
Example: construction projects	Example: R&D, software development

6.4 Worked Example

Activity with: Optimistic (a) = 2, Most Likely (m) = 5, Pessimistic (b) = 14

$$t_e = (2 + 4 \times 5 + 14) \div 6 = (2 + 20 + 14) \div 6 = 36 \div 6 = 6 \text{ days}$$

$$\sigma = (14 - 2) \div 6 = 12 \div 6 = 2 \text{ days}$$

$$\sigma^2 = 2^2 = 4$$

7. Cost of a Project

Project costs must be carefully estimated, budgeted, and controlled. Costs are broadly categorised as direct costs and indirect costs.

7.1 Types of Project Costs

Cost Type	Description	Examples
Direct Costs	Directly traceable to a specific activity or task	Labour, materials, equipment rental
Indirect Costs	Cannot be traced to a specific activity; shared overhead	Admin, utilities, management salaries
Fixed Costs	Do not change with the amount of work done	Rent, insurance, equipment purchase
Variable Costs	Change in proportion to the amount of work done	Hourly wages, materials used
Sunk Costs	Already incurred; cannot be recovered	Money already spent on cancelled phase
Opportunity Costs	Benefit foregone by choosing one option over another	Return on alternative investment

7.2 Cost-Time Trade-off (Crashing)

Crashing means reducing activity duration by spending more money (additional resources).

$$\text{Cost Slope} = (\text{Crash Cost} - \text{Normal Cost}) \div (\text{Normal Time} - \text{Crash Time})$$

Always crash the activity on the critical path with the LOWEST cost slope first.

7.3 Project Budget Components

- Labour costs (wages, salaries, benefits)
- Material costs (raw materials, components)
- Equipment costs (purchase or rental)
- Overhead costs (facilities, utilities)
- Contingency reserve (typically 10–15% of total budget)

8. Payback Period (PP)

The Payback Period is the time it takes for a project to recover its initial investment from net cash inflows. It is the simplest capital budgeting technique.

8.1 Formula

If cash flows are **EQUAL** each year:

$$PP = \text{Initial Investment} \div \text{Annual Cash Inflow}$$

If cash flows are **UNEQUAL**:

Add up cumulative cash inflows year by year until the investment is recovered.

$$PP = \text{Year before full recovery} + (\text{Remaining amount} \div \text{Cash flow in recovery year})$$

8.2 Decision Rule

- Accept if: $PP \leq \text{Target Payback Period}$
- Reject if: $PP > \text{Target Payback Period}$
- When comparing projects: Choose the one with the **SHORTER** payback period

8.3 Advantages & Disadvantages

Advantages	Disadvantages
Simple to calculate and understand	Ignores time value of money
Useful for liquidity assessment	Ignores cash flows AFTER the payback period
Helps assess risk (shorter PP = less risk)	Does not measure profitability
Good for companies with cash flow problems	Biased against long-term projects

8.4 Worked Example

Initial investment: R100,000

Year	Annual Cash Flow	Cumulative Cash Flow
1	R30,000	R30,000
2	R40,000	R70,000
3	R40,000	R110,000
4	R30,000	R140,000

Investment recovered during Year 3. Remaining after Year 2 = R100,000 – R70,000 = R30,000

$$PP = 2 + (30,000 \div 40,000) = 2 + 0.75 = 2.75 \text{ years} = 2 \text{ years and 9 months}$$

9. Accounting Rate of Return (ARR)

ARR (also called Return on Investment or Book Rate of Return) measures the average annual accounting profit as a percentage of the investment. It uses profit (not cash flow) and does not consider the time value of money.

9.1 Formula

$$\text{ARR} = (\text{Average Annual Profit} \div \text{Initial Investment}) \times 100$$

OR sometimes:

$$\text{ARR} = (\text{Average Annual Profit} \div \text{Average Investment}) \times 100$$

$$\text{Average Investment} = (\text{Initial Investment} + \text{Scrap Value}) \div 2$$

9.2 Calculating Average Annual Profit

$$\text{Annual Depreciation} = (\text{Initial Cost} - \text{Scrap Value}) \div \text{Useful Life}$$

$$\text{Average Annual Profit} = \text{Average Annual Cash Flow} - \text{Annual Depreciation}$$

9.3 Decision Rule

- Accept if: $\text{ARR} \geq \text{Target/Required Rate of Return}$
- Reject if: $\text{ARR} < \text{Target/Required Rate of Return}$
- When comparing: Choose project with HIGHER ARR

9.4 Advantages & Disadvantages

Advantages	Disadvantages
Simple to understand	Ignores time value of money
Based on accounting profit (familiar)	Uses profit not cash flow
Considers entire life of project	Ignores timing of profits
Relates to ROCE used by managers	Different ARR formulas cause confusion

9.5 Worked Example

Initial Investment: R200,000 | Scrap Value: R20,000 | Life: 5 years | Total Cash Flows: R350,000

$$\text{Average Annual Cash Flow} = \text{R}350,000 \div 5 = \text{R}70,000$$

$$\text{Annual Depreciation} = (200,000 - 20,000) \div 5 = \text{R}36,000$$

$$\text{Average Annual Profit} = \text{R}70,000 - \text{R}36,000 = \text{R}34,000$$

$$\text{ARR} = (\text{R}34,000 \div \text{R}200,000) \times 100 = 17\%$$

10. Net Present Value (NPV)

NPV is the difference between the present value of cash inflows and the present value of cash outflows over a period of time. It accounts for the time value of money (money today is worth more than money in the future).

10.1 Core Concept: Time Value of Money

R1,000 received today is worth more than R1,000 received in 3 years because today's money can be invested to earn a return. NPV discounts future cash flows back to today's value using a discount rate.

10.2 Formula

$$NPV = \Sigma [\text{Cash Flow} \div (1 + r)^t] - \text{Initial Investment}$$

Where: r = discount rate (cost of capital), t = time period (year)

Discount Factor Formula:

$$\text{Discount Factor} = 1 \div (1 + r)^t$$

(Discount factor tables are provided in most exams)

10.3 Decision Rule

- Accept if: $NPV > 0$ (project adds value)
- Reject if: $NPV < 0$ (project destroys value)
- If $NPV = 0$: Project just breaks even in present value terms
- When comparing projects: Choose the one with the HIGHEST positive NPV

10.4 Advantages & Disadvantages

Advantages	Disadvantages
Considers time value of money	Requires estimation of discount rate
Considers ALL cash flows over the project's life	Complex to calculate and explain
Gives absolute value (in Rands)	Difficult to compare projects of different sizes
Best method for maximising shareholder wealth	Sensitive to changes in discount rate

10.5 Worked Example

Investment: R50,000 | Discount Rate: 10% | Cash flows below:

Year	Cash Flow	Discount Factor (10%)	Present Value
1	R20,000	0.909	R18,180
2	R20,000	0.826	R16,520
3	R15,000	0.751	R11,265
Total PV			R45,965

$$NPV = R45,965 - R50,000 = -R4,035$$

NPV is NEGATIVE → REJECT the project (it destroys value at 10% discount rate)

11. Internal Rate of Return (IRR)

The IRR is the discount rate at which the NPV of a project equals ZERO. It represents the effective annual return the project generates. It is also called the breakeven discount rate.

11.1 Decision Rule

- Accept if: $IRR \geq \text{Required Rate of Return (cost of capital)}$
- Reject if: $IRR < \text{Required Rate of Return}$
- When comparing: Choose project with HIGHER IRR

11.2 How to Calculate IRR (Interpolation Method)

Step 1: Calculate NPV at a low rate (r_1) – should give a POSITIVE NPV

Step 2: Calculate NPV at a high rate (r_2) – should give a NEGATIVE NPV

Step 3: Use the interpolation formula:

$$IRR = r_1 + [NPV_1 \div (NPV_1 - NPV_2)] \times (r_2 - r_1)$$

Note: NPV1 is positive; NPV2 is negative (use their absolute values carefully)

11.3 Advantages & Disadvantages

Advantages	Disadvantages
Considers time value of money	Can give multiple IRRs if cash flows change sign
Considers all cash flows	Cannot rank mutually exclusive projects reliably
Easy to compare with cost of capital	Assumes reinvestment at IRR (not always realistic)
Expressed as a percentage (intuitive)	More complex to calculate than PP or ARR

11.4 Worked Example

Initial Investment: R100,000

Annual Cash Flow: R30,000 for 5 years

Required rate: 12%

Try $r_1 = 10\%$: $NPV_1 = R30,000 \times 3.791 - R100,000 = +R13,730$

Try $r_2 = 20\%$: $NPV_2 = R30,000 \times 2.991 - R100,000 = -R10,270$

$$IRR = 10\% + [13,730 \div (13,730 + 10,270)] \times (20\% - 10\%)$$

$$IRR = 10\% + [13,730 \div 24,000] \times 10\% = 10\% + 5.72\% \approx 15.72\%$$

IRR (15.72%) > Required Rate (12%) → ACCEPT the project

11.5 Summary: Capital Budgeting Methods Comparison

Method	Uses TVM?	Uses Profit or CF?	Decision Rule	Best For
PP	No	Cash Flow	Shortest PP	Liquidity / Risk
ARR	No	Profit	Highest ARR	Accounting reports
NPV	Yes	Cash Flow	Highest positive NPV	Wealth maximisation
IRR	Yes	Cash Flow	IRR $\geq$ Required Rate	% return comparison

PART 2: EXAM PRACTICE WITH ANSWERS

Section A: Management Theory Questions

Q1: Define management in your own words and name the four key resources managers work with.

Answer: Management is the process of coordinating people and resources to achieve organisational goals efficiently and effectively. The four key resources are: (1) Human resources (people), (2) Financial resources (money), (3) Physical resources (materials, equipment), (4) Informational resources (data and knowledge).

Q2: Explain the difference between efficiency and effectiveness in management.

Answer: Efficiency means doing things right – achieving maximum output with minimum input (no waste). Effectiveness means doing the right things – achieving the intended goals and outcomes. A manager should strive to be BOTH efficient AND effective. Example: A factory can be efficient (low cost per unit) but ineffective (producing products nobody wants to buy).

Q3: List and explain Fayol's 5 functions of management.

Answer: (1) Planning: Setting objectives and deciding on strategies to achieve them. (2) Organising: Arranging resources and tasks to meet goals, creating the organisational structure. (3) Commanding: Directing employees to perform their duties. (4) Coordinating: Ensuring all departments and activities work together harmoniously. (5) Controlling: Measuring actual performance against set targets and correcting deviations.

Q4: Explain the principle of Unity of Command and why it is important.

Answer: Unity of Command states that each employee should receive orders from only ONE superior. This avoids confusion, conflicting instructions, and divided loyalties. If an employee has two bosses who give different instructions, the employee will not know which to follow, leading to chaos and poor performance.

Q5: Compare Fayol and Mintzberg's approaches to understanding management.

Answer: Fayol's approach is PRESCRIPTIVE: he says what managers SHOULD do (plan, organise, command, coordinate, control). It is systematic and function-based. Mintzberg's approach is DESCRIPTIVE: he says what managers ACTUALLY do based on observation. He found managers do 10 roles that are fragmented and varied. Mintzberg criticised Fayol for being too simplistic and not reflecting reality.

Q6: Mintzberg identified three categories of managerial roles. Name them and give one role from each.

Answer: (1) Interpersonal roles – involve interactions with other people. Example: Leader (motivating and directing staff). (2) Informational roles – involve processing and sharing information. Example: Monitor (gathering internal and external information). (3) Decisional roles – involve making decisions. Example: Entrepreneur (initiating innovation and change).

Q7: Give an example of a manager acting in the 'Disturbance Handler' role.

Answer: A disturbance handler deals with unexpected crises. Example: A factory manager who immediately responds when a key machine breaks down during production – they reorganise work shifts, call in a repair technician, and communicate with clients about possible delays. Another example: a manager resolving a sudden strike action by union members.

Section B: Project Management Questions

Q8: Define a 'project' and list THREE characteristics that distinguish it from routine operations.

Answer: A project is a temporary, unique endeavour with a defined start and end, aimed at achieving specific objectives within constraints of scope, time, and cost. Three characteristics: (1) Temporary – has a definite end date. (2) Unique – the output has never been produced in exactly the same way before. (3) Progressive elaboration – it develops in steps, with increasing detail over time.

Q9: Explain the Triple Constraint in project management.

Answer: The Triple Constraint (Project Triangle) states that every project is limited by three interrelated factors: (1) Scope – what work must be done. (2) Time – how long the project will take. (3) Cost – how much budget is available. Changing one constraint affects the others. If you increase scope, time or cost must increase too. Quality sits at the centre of the triangle.

Section C: CPM & PERT Calculation Questions

Question 10: CPM Network Analysis

Given the following activities, draw a network diagram and find: (a) the earliest and latest times, (b) the float for each activity, (c) the critical path and project duration.

Activity	Predecessor(s)	Duration (days)
A	–	4
B	–	6
C	A	3
D	A, B	5
E	C	4
F	D, E	2

ANSWER:

Activity	Duration	EST	EFT	LST	LFT	Float	Critical?
A	4	0	4	0	4	0	YES
B	6	0	6	2	8	2	No
C	3	4	7	4	7	0	YES
D	5	6	11	8	13	2	No
E	4	7	11	7	11	0	YES
F	2	11	13	11	13	0	YES

Critical Path: A → C → E → F = 4+3+4+2 = 13 days

Question 11: PERT Calculation

For each activity, calculate the Expected Time (t_e) and Variance:

Activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	1	3	5
B	2	5	14
C	3	4	7

ANSWER:

Activity	$t_e = (a+4m+b)÷6$	Variance = $[(b-a)÷6]^2$
A	$(1+12+5)÷6 = 18÷6 = 3$ days	$[(5-1)÷6]^2 = (0.67)^2 = 0.44$
B	$(2+20+14)÷6 = 36÷6 = 6$ days	$[(14-2)÷6]^2 = (2)^2 = 4$
C	$(3+16+7)÷6 = 26÷6 = 4.33$ days	$[(7-3)÷6]^2 = (0.67)^2 = 0.44$

Section D: Financial Appraisal Questions

Question 12: Payback Period

A company is considering two projects. Each requires an initial investment of R150,000. Calculate the Payback Period for each and state which should be chosen.

Year	Project A Cash Flow	Project B Cash Flow
1	R40,000	R70,000
2	R50,000	R50,000
3	R60,000	R40,000
4	R50,000	R30,000

ANSWER:

Project A cumulative: Y1=40k, Y2=90k, Y3=150k → PP = exactly 3 years

Project B cumulative: Y1=70k, Y2=120k, Y3=160k → Remaining after Y2 = 30k; PP = 2 + (30,000/40,000) = 2.75 years

Choose Project B – Payback Period of 2.75 years vs 3 years for Project A

Question 13: ARR Calculation

Investment: R300,000 | Scrap Value: R30,000 | Useful Life: 6 years | Total Cash Inflows: R480,000 | Required ARR: 15%

Calculate the ARR and state whether the project should be accepted.

ANSWER:

Average Annual Cash Flow = R480,000 ÷ 6 = R80,000

Annual Depreciation = (R300,000 – R30,000) ÷ 6 = R45,000

Average Annual Profit = R80,000 – R45,000 = R35,000

$$\text{ARR} = (\text{R}35,000 \div \text{R}300,000) \times 100 = 11.67\%$$

ARR (11.67%) < Required Rate (15%) → REJECT the project

Question 14: NPV Calculation

Initial Investment: R80,000 | Discount Rate: 12% | Cash flows:

Year 1: R25,000 | Year 2: R30,000 | Year 3: R35,000 | Year 4: R20,000

Discount factors at 12%: Y1=0.893, Y2=0.797, Y3=0.712, Y4=0.636

Should this project be accepted?

ANSWER:

Year	Cash Flow	Discount Factor	Present Value
1	R25,000	0.893	R22,325

Year	Cash Flow	Discount Factor	Present Value
2	R30,000	0.797	R23,910
3	R35,000	0.712	R24,920
4	R20,000	0.636	R12,720
Total PV			R83,875

$$NPV = R83,875 - R80,000 = +R3,875$$

NPV is POSITIVE (→ R3,875) → ACCEPT the project

Question 15: IRR Calculation

Initial Investment: R120,000 | Required Rate: 14%

Annual Cash Flow: R35,000 for 5 years

Discount factors: At 10%: 3.791 | At 18%: 3.127

Calculate the IRR and state the decision.

ANSWER:

NPV at $r_1=10\%$: $R35,000 \times 3.791 - R120,000 = R132,685 - R120,000 = +R12,685$

NPV at $r_2=18\%$: $R35,000 \times 3.127 - R120,000 = R109,445 - R120,000 = -R10,555$

$$IRR = 10\% + [12,685 \div (12,685 + 10,555)] \times (18\% - 10\%)$$

$$IRR = 10\% + [12,685 \div 23,240] \times 8\% = 10\% + 4.37\% = 14.37\%$$

IRR (14.37%) > Required Rate (14%) → ACCEPT the project (marginally)

EXAM QUICK REFERENCE – FORMULA CHEAT SHEET

Topic	Formula / Key Fact
PERT Expected Time	$t_e = (a + 4m + b) \div 6$
PERT Variance	$\sigma^2 = [(b - a) \div 6]^2$
CPM Forward Pass	EFT = EST + Duration; EST at node = max EFT of predecessors
CPM Backward Pass	LST = LFT – Duration; LFT at node = min LST of successors
CPM Float	Float = LFT – EFT (or LST – EST); Critical path has Float = 0
Payback Period (equal CF)	PP = Initial Investment $\div$ Annual Cash Flow
Payback Period (unequal CF)	PP = Year before + (Remaining $\div$ Next year CF)
ARR	ARR = (Avg Annual Profit $\div$ Initial Investment) $\times$ 100
Average Annual Profit	Avg Annual CF – Annual Depreciation
Annual Depreciation	(Cost – Scrap Value) $\div$ Useful Life
NPV	NPV = $\Sigma[CF \div (1+r)^t]$ – Initial Investment
Discount Factor	$1 \div (1 + r)^t$
IRR Interpolation	IRR = $r_1 + [NPV_1 \div (NPV_1 + NPV_2)] \times (r_2 - r_1)$
Cost Slope (Crashing)	(Crash Cost – Normal Cost) $\div$ (Normal Time – Crash Time)

Method	Accept When
Payback Period	PP $\leq$ Target Period
ARR	ARR $\geq$ Required Rate
NPV	NPV > 0
IRR	IRR $\geq$ Cost of Capital